

Luca Giannotti

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EDUCATION

Texas A&M University

College Station, TX

Bachelor of Science (Hons.) in Computer Science, Minor in Philosophy & Mathematics

Aug. 2024 – May 2028

- **Relevant Coursework:** Program Design & Concepts, Data Structures & Algorithms, Computer Organization/Architecture, Programming Languages, Linear Algebra, Vector/Multivariable Calculus, Foundations of Software Engineering, Design and Analysis of Algorithms, Computer Systems
- **Extracurricular:** Aggie Coding Club, Texas A&M Computing Society, Texas A&M Baseball R&D, STMI Lab

EXPERIENCE

Undergraduate Researcher

August 2026 – Present

STMI Lab (Systems, Technologies & Machine Intelligence), Texas A&M CSE Dept.

College Station, TX

- Research macronutrient sensing using deep learning, applying time-series and glucose foundation models to improve nutrition estimation from sensor data.
- Collaborate with Dr. McShane's group (TAMU Biomedical Engineering) to extend the approach to protein estimation using data from a newly developed sensor.
- Secured a Texas A&M HPRC computing allocation to support model training on large-scale time-series data.

Data Analyst Intern

June 2026 – August 2026

Indigenous Pact

- Led a comprehensive data quality audit of 52 gold-layer tables in a Databricks/Azure lakehouse environment during a large-scale data migration, using SQL and PySpark to validate schema integrity and surface data reliability issues.
- Developed a repeatable validation methodology to cross-check migrated datasets against other records and intended output, identifying and documenting 150+ discrete data integrity issues.
- Synthesized audit findings into structured Power BI reporting artifacts, translating raw technical findings into organized, interpretable documentation.

PROJECTS

TAMU Statistics – tamustatistics.com | *Next.js, FastAPI, Python, Pandas, React, Docker* Mar 2025 – Apr 2025

- Developed and deployed an interactive data analysis web app processing 1,000+ GPA and course datasets for Texas A&M University.
- Built backend automation pipelines using Pandas, PyPDF, Requests, and Selenium to scrape, clean, and structure academic datasets.
- Designed reproducible data workflows and containerized deployment using Docker.

Homelab Infrastructure | *Proxmox, Linux, Docker, YAML, Nginx, Cloudflare*

May 2025 – Present

- Built and maintain a self-hosted Proxmox homelab running 10+ containerized services for media automation, cloud access, websites, and software projects.
- Implemented Nginx reverse proxy and Cloudflare Tunnels for secure external access and zero-trust remote connectivity.
- Automated service orchestration and configuration using Docker Compose and YAML-based workflows.

TECHNICAL SKILLS

Languages: C/C++, Python, SQL (Postgres), JavaScript, HTML/CSS, Bash

Frameworks: React, Node.js, Flask, FastAPI

Developer Tools: Git, Docker, VS Code, PyCharm, WebStorm, Linux/Unix & Windows, WSL, Proxmox, Networking, Power BI, Azure, Databricks